Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[18ME52]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

APPLIED THERMODYNAMICS

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

iii. Use of Steam tables and thermodynamics data hand book is permitted

1. a. With the help of P-v and T-s diagrams, Obtain an expression for air standard efficiency of diesel cycle in terms of compression ratio and cut-off ratio. **07**

b. A boiler produces dry and saturated steam at 30 bar. The steam expands in the turbine to a condenser pressure of 20 kPa. Compare the cyclic work done and thermal efficiency of Carnot and Rankine cycles for these conditions. **07**

2. a. With the help of a neat sketch, explain the working of battery coil ignition system. **06**

b. During the trial of single-cylinder, four stroke oil engine, the following results were obtained. **08**

Cylinder diameter = 20 cm

Stroke = 40 cm

Mean effective pressure = 6 bar

Torque = 407 N-m ; Speed = 250 rpm

Oil consumption = 4 kg/h

Calorific value of fuel = 43 MJ/kg

Cooling water flow rate = 4.5 kg/min

Air used per kg of fuel = 30 kg

Rise in cooling water temperature = 45°C

Temperature of exhaust gases = 420°C

Room temperature = 20°C

Mean specific heat of exhaust gas = 1 kJ/kg-K

Specific heat of water = 4.18 kJ/kg-K

Find the IP, BP and draw up a heat balance sheet for the test in kJ/h

3. a. Derive an expression for work done with clearance volume for reciprocating air compressor. **06**

b. A single-stage double acting reciprocating air compressor takes in 14 m^3 of air per minute measured at 1.013 bar and 15°C. The delivery pressure is 7 bar and the compressor speed is 300 rpm. The compressor has a clearance volume of 5% of swept volume with a compression and re-expansion index of n = 1.3. **08**

Calculate the swept volume of the cylinder, the delivery temperature and the indicated power.

OR

4. **a.** Deduce the condition for minimum power in a two-stage compressor with perfect intercooling. **07**

b. With a neat sketch, explain the working principle of popularly used rotary compressor. **07**

5. **a.** Obtain an expression for critical pressure ratio for maximum discharge through the nozzle **07**

b. Calculate the critical pressure and throat area per unit mass-flow rate of steam, expanding through a convergent-divergent nozzle from 10 bar, dry saturated, down to atmospheric pressure of 1 bar. Assume that the inlet velocity is negligible and that the expansion is isentropic. **07**

OR

6. **a.** What is compounding and what is the need of compounding? Explain pressure-velocity compounding with neat sketch. **06**

b. The steam is supplied to a De-Laval turbine at a velocity of 1000 m/s at an angle of 20°. The blade velocity is 300 m/s and the blades are symmetrical. The mass-flow rate of the steam is 0.5 kg/s. Allowing a friction factor of 0.8, determine. **08**

i. Blade efficiency

ii. Power developed

iii. Stage efficiency, if the nozzle efficiency is 95%

7. **a.** With the help of P-h diagram, explain the effect of following conditions on the performance of vapour compression refrigeration cycle: **08**

i. Effect of evaporator pressure

ii. Effect of subcooling

b. Explain the working of vapour absorption refrigeration system with the help of neat sketch. **06**

OR

8. **a.** With the help of a neat sketch, explain the working of winter air conditioning system. **08**

b. Discuss about the desirable properties of refridgerant. **06**